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Prediction of performance of simple gas turbines

From cycle calculations such as those of Chapter 2, it is possible to determine the pressure ratio which for any given maximum cycle temperature will give the greatest overall efficiency, and the mass flow required to give the desired power. When such preliminary calculations have been made, the most suitable design data for any particular application may be chosen. It is then possible to design the individual components of a gas turbine so that the complete unit will give the required performance when running at the *design point*; that is, when it is running at the particular speed, pressure ratio and mass flow for which the components were designed. The problem then remains to find the variation of performance of the gas turbine over the complete operating range of speed and power output, which is normally referred to as *off-design performance*.

The performance characteristics of the individual components may be estimated on the basis of previous experience or obtained from actual tests. When the components are linked together in an engine, the range of possible operating conditions for each component is considerably reduced. The problem is to find corresponding operating points on the characteristics of each component when the engine is running at a steady speed, or in *equilibrium* as it is frequently termed. The equilibrium running points for a series of speeds may be plotted on the compressor characteristic and joined up to form an *equilibrium running line* (or *zone*, depending upon the type of gas turbine and load), the whole forming an *equilibrium running diagram*. Once the operating conditions have been determined, it is a relatively simple matter to obtain performance curves of power output or thrust, and specific fuel consumption.

The equilibrium running diagram also shows the proximity of the operating line or zone to the compressor surge line. If it intersects the surge line the gas turbine will not be capable of being brought up to full speed without some remedial action being taken. It is this phenomenon which was referred to when speaking of 'stability of operation' in the Introduction. Finally, it shows whether the engine is operating in a region of adequate compressor efficiency; ideally the operating line or zone should lie near the locus of points of maximum compressor efficiency shown in Fig. 4.10(a).

The variation of specific fuel consumption with reduction in power, sometimes referred to as *part-load performance*, is of major importance in applications where considerable running at low power settings is required. This would be the case for

any vehicular gas turbine, and the poor specific fuel consumption at part load is probably the biggest disadvantage of the gas turbine for vehicular use. The fuel consumption of aircraft gas turbines at reduced power is of critical importance when extensive delays occur either at the point of departure or at the destination. In one case the idling fuel consumption when taxiing is important, and in the other the fuel flow at low flight speeds and medium altitudes is critical.

When determining the off-design performance it is important to be able to predict not only the effect on specific fuel consumption of operation at part load, but also the effect of ambient conditions on maximum output. The effects of high and low ambient temperatures and pressures must all be considered. Land-based gas turbines may operate between ambient temperatures of -60°C in the Arctic and 50°C in the tropics, and at altitudes from sea level to about 3000 metres, while aircraft gas turbines have to operate over much wider ranges of inlet temperature and pressure. The variation of maximum power with ambient conditions is clearly of prime importance to the customer, and the manufacturer must be prepared to guarantee the performance available at any specified condition. If, for example, we consider the use of gas turbines for peak-load generation of electricity, the peak loads will occur in the coldest days of winter in Europe, but are much more likely to occur in the summer in the USA because of the heavy demand for air-conditioning and refrigeration systems. The effect of ambient conditions on the performance of aircraft gas turbines has a critical effect on the runway length required and the payload that can be carried, and hence on both safety and economics.

The basic methods for determining equilibrium off-design performance of simple gas turbines will be described here, while more complex arrangements and transient operation are dealt with in Chapter 10. The types of gas turbine discussed in this chapter will be (a) the single-shaft unit delivering shaft power, (b) the free turbine engine, where the gas-generator turbine drives the compressor and the power turbine drives the load, and (c) the simple jet engine where the useful output is a high-velocity jet produced in the propelling nozzle. Schematics of these layouts are shown in Fig. 9.1, and it can readily be observed that the gas generator performs exactly the same function for both the free turbine engine and the jet engine. The flow characteristics of a free turbine and a nozzle are similar, and they impose the same restrictions on the operation of the gas generator, with the result that the free turbine engine and the jet engine are thermodynamically similar and differ only in the manner in which the output is utilized. Several successful jet engines have been converted to shaft power use by substituting a free power turbine for the propelling nozzle, and this approach has been widely used for pipeline compression, marine propulsion and electricity generation. An important practical result of the flow equivalency of power turbines and propelling nozzles is that overhaul shops can test gas generators on a jet engine test bed, without the need for a complex and expensive dynamometer. In pipeline and marine applications, overhaul is normally done by replacing the gas generator while leaving the power turbine *in situ*; it is common to replace a gas generator within a 24-hour period, and some ships actually carry a spare gas generator. Pipelines often own several extra gas generators, which can be installed to meet overhaul requirements at different sites.

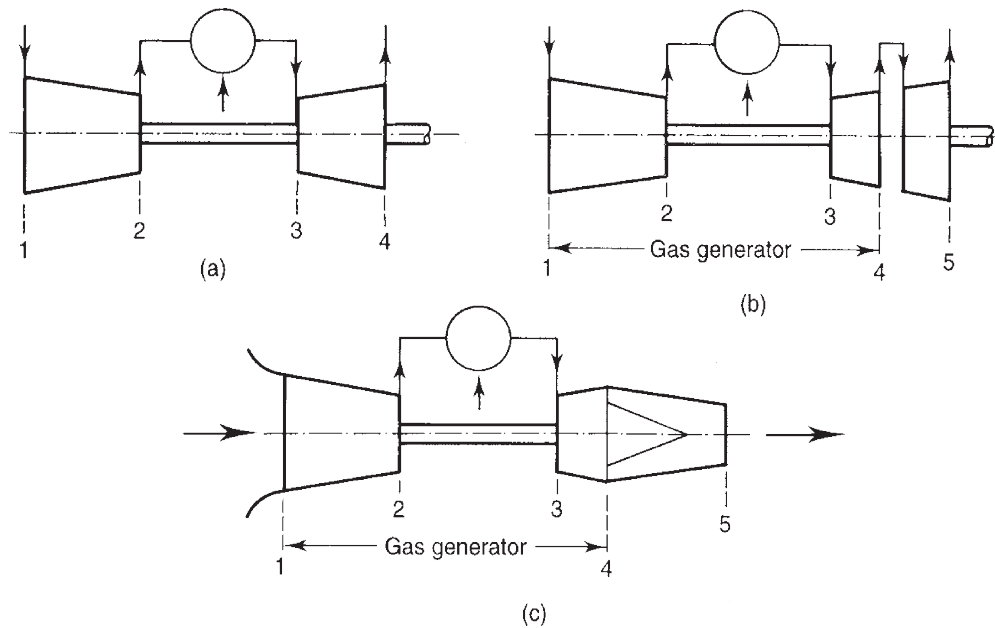


FIG. 9.1 Simple gas turbine units

All off-design calculations depend on satisfying the essential conditions of *compatibility of mass flow, work and rotational speed* between the various components; these are often referred to as *matching calculations*. The basic methods suitable for hand calculations will be presented in detail; obviously in industry these calculations would be done using sophisticated software, but it is very important that performance engineers fully understand the thermodynamic basis of the matching procedures.

It is logical to deal with the single-shaft engine first, being the simplest to analyse, and then proceed to the free turbine engine, which introduces the added complicity of flow compatibility between the gas-generator turbine and the power turbine. Lastly, the chapter will close with the jet engine where there is the further complication of forward speed and altitude effects. Analysis of multi-spool configurations will be dealt with in the next chapter.

9.1 Component characteristics

The variation of mass flow, pressure ratio and efficiency with rotational speed of the compressor and turbine is obtained from the compressor and turbine characteristics, examples of which are given in Chapters 4, 5 and 7. It is convenient to represent the compressor characteristic as shown in Fig. 9.2, with the variation of efficiency along each constant speed line plotted versus both mass flow and pressure ratio. With high-performance axial compressors the constant speed lines become vertical on a mass flow basis when the inlet is choked, and in this region it is essential to plot the efficiency as a function of pressure ratio. The turbine

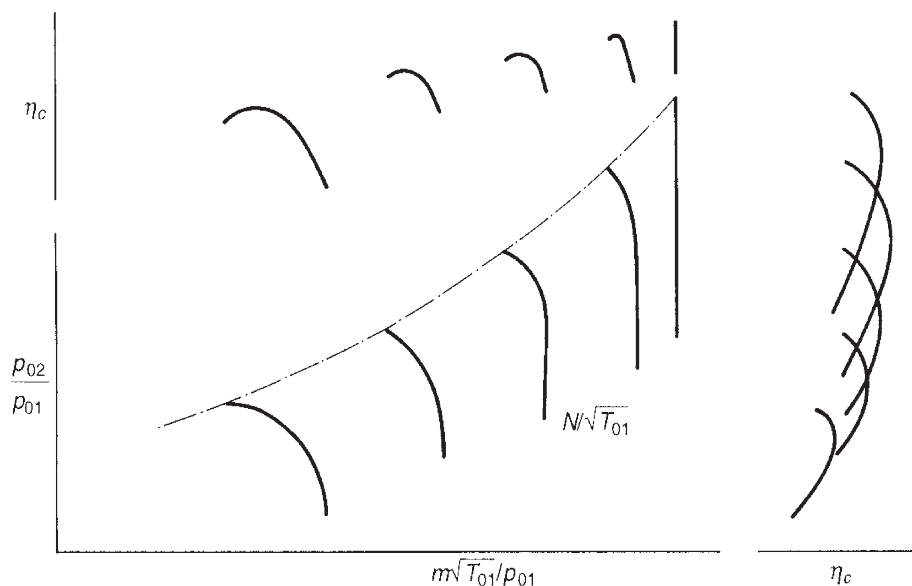
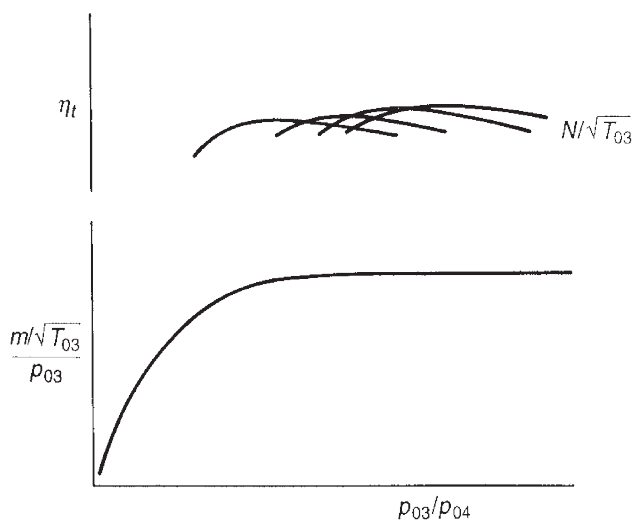


FIG. 9.2 Compressor characteristics

characteristic can be used in the form given in Fig. 7.24. It is often found in practice, however, that turbines do not exhibit any significant variation in non-dimensional flow with non-dimensional speed, and in most cases the turbine operating region is severely restricted by another component downstream of it. In explaining the method used for off-design performance calculations it will initially be assumed that the mass flow function can be represented by a single curve as in Fig. 9.3. The modification necessary to account for a family of constant speed curves will be discussed in section 9.3.



Note: parameters will be $\frac{m\sqrt{T_{04}}}{p_{04}}$, $\frac{p_{04}}{p_a}$, $\frac{N_p}{\sqrt{T_{04}}}$ and η_{tp} for a power turbine

FIG. 9.3 Turbine characteristics

For accurate calculations it is necessary to consider the variation of pressure losses in the inlet ducting, the combustion chamber and the exhaust ducting. These are essentially secondary effects, however, and the off-design calculations will be introduced on the basis of negligible inlet and exhaust losses and a combustion chamber pressure loss which is a fixed percentage of the compressor delivery pressure. Such approximations are quite adequate for many purposes. For detailed calculations it would be normal to use a digital computer, when methods of allowing for variable pressure losses are easily introduced. Further discussion of this matter will be deferred until section 9.7.

9.2 Off-design operation of the single-shaft gas turbine

Referring to the single-shaft gas turbine shown in Fig. 9.1(a), it can readily be seen that when inlet and exhaust pressure losses are ignored, the pressure ratio across the turbine is determined by the compressor pressure ratio and the pressure loss in the combustion chamber. The mass flow through the turbine will be equal to the mass flow delivered by the compressor, less any bleeds, but supplemented by the fuel flow; it has been pointed out earlier that the bleeds are often approximately equal to the fuel flow. In general terms, the procedure for obtaining an equilibrium running point is as follows:

- (i) Select a constant speed line on the compressor characteristic and choose any point on this line; the values of $m\sqrt{T_{01}}/p_{01}$, p_{02}/p_{01} , η_c and $N\sqrt{T_{01}}$ are then determined.
- (ii) The corresponding point on the turbine characteristic is obtained from consideration of compatibility of rotational speed and flow.
- (iii) Having matched the compressor and turbine characteristics, it is necessary to ascertain whether the work output corresponding to the selected operating point is compatible with that required by the driven load; this requires knowledge of the variation of power with speed, which depends on the manner in which the power is absorbed.

The compressor and turbine are directly coupled together, so that compatibility of rotational speed requires

$$\frac{N}{\sqrt{T_{03}}} = \frac{N}{\sqrt{T_{01}}} \times \sqrt{\frac{T_{01}}{T_{03}}} \quad (9.1)$$

Compatibility of flow between the compressor and turbine can be expressed in terms of the non-dimensional flows by the identity

$$\frac{m_3\sqrt{T_{03}}}{p_{03}} = \frac{m_1\sqrt{T_{01}}}{p_{01}} \times \frac{p_{01}}{p_{02}} \times \frac{p_{02}}{p_{03}} \times \sqrt{\frac{T_{03}}{T_{01}}} \times \frac{m_3}{m_1}$$

The pressure ratio p_{03}/p_{02} can be obtained directly from the combustion pressure loss, i.e. $p_{03}/p_{02} = 1 - (\Delta p_b/p_{02})$. It will normally be assumed that

$m_1 = m_3 = m$, but variation in mass flow at different points in the engine can easily be included if required. Rewriting the previous equation in terms of m , we get

$$\frac{m\sqrt{T_{03}}}{p_{03}} = \frac{m\sqrt{T_{01}}}{p_{01}} \times \frac{p_{01}}{p_{02}} \times \frac{p_{02}}{p_{03}} \times \sqrt{\frac{T_{03}}{T_{01}}} \quad (9.2)$$

Now $m\sqrt{T_{01}}/p_{01}$ and p_{02}/p_{01} are fixed by the chosen operating point on the compressor characteristic, p_{03}/p_{02} is assumed to be constant and $m\sqrt{T_{03}}/p_{03}$ is a function of the turbine pressure ratio p_{03}/p_{04} . Neglecting inlet and exhaust pressure losses, $p_a = p_{01} = p_{04}$, so that the turbine pressure ratio can be calculated from $p_{03}/p_{04} = (p_{03}/p_{02})(p_{02}/p_{01})$. Thus all the terms of equation (9.2) with the exception of $\sqrt{(T_{03}/T_{01})}$ can be obtained from the compressor and turbine characteristics. The turbine inlet temperature T_{03} can therefore be obtained from equation (9.2) when the ambient temperature, which is equal to T_{01} , is specified.

Having determined the turbine inlet temperature, the turbine non-dimensional speed $N/\sqrt{T_{03}}$ is obtained from equation (9.1). The turbine efficiency can then be obtained from the turbine characteristic using the known values of $N/\sqrt{T_{03}}$ and p_{03}/p_{04} , and the turbine temperature drop can be calculated from

$$\Delta T_{034} = \eta_t T_{03} \left[1 - \left(\frac{1}{p_{03}/p_{04}} \right)^{(\gamma-1)/\gamma} \right] \quad (9.3)$$

The compressor temperature rise for the point selected on the compressor characteristic can be similarly calculated as

$$\Delta T_{012} = \frac{T_{01}}{\eta_c} \left[\left(\frac{p_{02}}{p_{01}} \right)^{(\gamma-1)/\gamma} - 1 \right] \quad (9.4)$$

The net power output corresponding to the selected operating point is then found from

$$\text{net power output} = mc_{pg}\Delta T_{034} - \frac{1}{\eta_m} mc_{pa}\Delta T_{012}^\dagger \quad (9.5)$$

where η_m is the mechanical efficiency of the compressor–turbine combination, and m is given by $(m\sqrt{T_{01}}/p_{01})(p_a/\sqrt{T_a})$ for prescribed ambient conditions.

Finally, it is necessary to consider the characteristics of the load to determine whether the compressor operating point selected represents a valid solution. If, for example, the engine were run on a test bed coupled to a hydraulic or electrical dynamometer, the load could be set independently of the speed and it would be possible to operate at any point on the compressor characteristic within the temperature limit imposed by safety considerations. With a fixed-pitch propeller load, however,

[†] Distinguishing suffixes will be added to c_p only in equations where both c_{pa} and c_{pg} appear simultaneously. In other cases it will be clear from the context which mean c_p should be used. γ will be treated similarly.

the power absorbed varies as the cube of the rotational speed of the propeller. When the transmission efficiency and gear ratio are known, the load characteristic in terms of the net power actually required from the turbine and the turbine speed can be plotted as in Fig. 9.4. The problem then becomes one of finding the *single* point on each constant speed line of the compressor characteristic which will give the required net power output at that speed; this can only be done by trial and error, taking several operating points on the compressor characteristic and establishing the power output corresponding to each one. If the calculated net power output for any point on the compressor characteristic is not equal to the power required at the selected speed, the engine will not be in equilibrium and will either accelerate or decelerate depending on whether there is a surplus or deficiency of power. Repeating this procedure for a series of constant speed lines, a series of points are obtained, which can be joined up to form the equilibrium running line as shown in Fig. 9.5.

The most common types of load used with single-shaft gas turbines are the electric generator and variable-pitch aircraft propellers, both of which run at a constant rotational speed with the load varied electrically or by changing pitch at different flight conditions. Thus the equilibrium lines for these loads would correspond to a particular line of constant non-dimensional speed, as shown in Fig. 9.5, and each point on this line would represent a different value of fuel flow, turbine inlet temperature and power output. At each speed it is possible to find, by trial and error, the compressor operating point corresponding to zero net output, and the no-load running line is also shown in Fig. 9.5.

The equilibrium running lines depicted show that a fixed-pitch propeller implies operation in a zone of high compressor efficiency over a wide range of output, whereas the generator and variable-pitch propeller result in a rapid drop in compressor efficiency as load is reduced. The location of the equilibrium running line relative to the surge line indicates whether the engine can be brought up to full power without any complications. The equilibrium running line for the fixed-pitch propeller lies close to the surge line, and may even intersect it, in which case the engine could not be accelerated to full power. This can be overcome by incorporating a

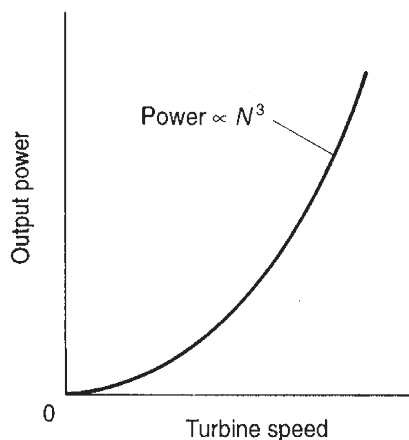


FIG. 9.4 Load characteristics

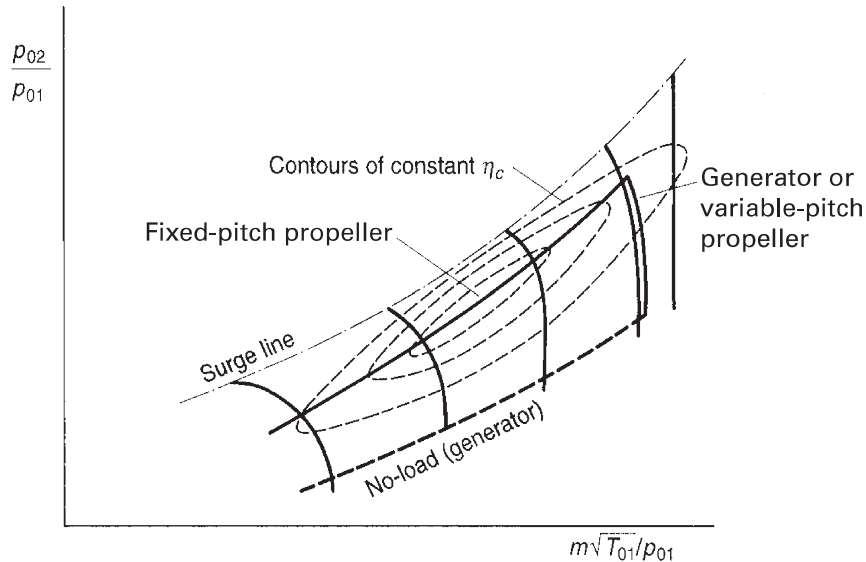


FIG. 9.5 Equilibrium running lines

blow-off valve towards the rear of the compressor. Section 9.6 deals with the matter in more detail. The running line for the generator or variable-pitch propeller can be seen to be well away from surge, and the gas turbine could be accelerated to full speed before applying the load without any surge problem being encountered.

The calculations described above determine the values of all the parameters required for a complete performance calculation for any point within the operating range. T_{03} is known, and T_{02} is found from $(\Delta T_{012} + T_{01})$. Thus the combustion temperature rise is known and it is possible to obtain the fuel/air ratio f from the curves of Fig. 2.17 and an assumed value of combustion efficiency. The fuel flow is then given by mf . From the fuel flow and power output at each operating point the variation in specific fuel consumption (or thermal efficiency) with load can be determined. The results refer to operation at the assumed value of T_{01} ($= T_a$) and p_{01} ($= p_a$), but the process could be repeated over the range of values of ambient temperature and pressure likely to be encountered.

The matching calculations for a single-shaft gas turbine are illustrated in the following example.

EXAMPLE 9.1

The following data refer to a single-shaft gas turbine operating at its design speed:

Compressor characteristic			Turbine characteristic	
p_{02}/p_{01}	$m\sqrt{T_{01}}/p_{01}$	η_c	$m\sqrt{T_{03}}/p_{03}$	η_t
5.0	329.0	0.84	139.0	0.87
4.5	339.0	0.79	(both constant over range of pressure ratio considered)	
4.0	342.0	0.75		

Assuming ambient conditions of 1.013 bar and 288 K, a mechanical efficiency of 98 per cent, and neglecting all pressure losses, calculate the turbine inlet temperature required for a power output of 3800 kW. The 'non-dimensional' flows are expressed in terms of kg/s, K and bar.

The method of solution is to establish, for each point given on the compressor characteristic, the turbine inlet temperature from equation (9.2), the compressor temperature rise from equation (9.4) and the turbine temperature drop from equation (9.3). Once these have been established, the power output can be obtained from equation (9.5); it is then necessary to plot turbine inlet temperature versus power output to find the required temperature for an output of 3800 kW.

Taking the compressor operating point at a pressure ratio of 5.0,

$$\sqrt{\frac{T_{03}}{T_{01}}} = \frac{(m\sqrt{T_{03}/p_{03}})(p_{03}/p_{01})}{m\sqrt{T_{01}/p_{01}}}$$

With pressure losses neglected, $p_{03} = p_{02}$, and hence

$$\sqrt{\frac{T_{03}}{T_{01}}} = \frac{139.0 \times 5.0}{329.0} = 2.11$$

giving $T_{03} = 1285$ K.

The compressor temperature rise is given by

$$\Delta T_{012} = \frac{288}{0.84}[5.0^{1/3.5} - 1] = 200.5 \text{ K}$$

The turbine temperature drop is given by

$$\Delta T_{034} = 0.87 \times 1285 \left[1 - \frac{1}{5.0^{1/4}} \right] = 370.0 \text{ K}$$

The air mass flow is obtained from the non-dimensional flow entering the compressor,

$$m = 329 \times \frac{1.013}{\sqrt{288}} = 19.64 \text{ kg/s}$$

The power output can now be obtained, giving

$$\begin{aligned} \text{Power output} &= 19.64 \times 1.148 \times 370.0 - \left(\frac{19.64 \times 1.005 \times 200.5}{0.98} \right) \\ &= 8340 - 4035 \\ &= 4305 \text{ kW} \end{aligned}$$

Thus for a power output of 4305 kW the turbine inlet temperature is 1285 K.

Repeating the calculation for the three points given on the compressor characteristic yields the following results:

p_{02}/p_{01}	$\frac{T_{03}}{[K]}$	$\frac{\Delta T_{012}}{[K]}$	$\frac{\Delta T_{034}}{[K]}$	$\frac{m}{[kg/s]}$	$\frac{\text{Power output}}{[kW]}$
5.0	1285	200.5	370.0	19.64	4305
4.5	982	196.1	267.0	20.25	2130
4.0	761	186.7	194.0	20.4	635

Plotting the value of T_{03} versus power output, it is found that for an output of 3800 kW the required turbine inlet temperature is 1215 K.

9.3 Equilibrium running of a gas generator

It was pointed out at the start of this chapter that the gas generator performs the same function for both the free turbine engine and the jet engine, i.e. the generation of a continuous flow of gas at high pressure and temperature which can be expanded to a lower pressure to produce either shaft work or a high-velocity propulsive jet. Before considering either type of engine, it is appropriate to consider the behaviour of the gas generator alone.

Considerations of compatibility of speed and flow are the same as for the single-shaft engine described in the previous section, and equations (9.1) and (9.2) are applicable. This time, however, the pressure ratio across the turbine is not known, and it must be determined by equating the turbine work to the compressor work. The required turbine temperature drop, in conjunction with the turbine inlet temperature and efficiency, determines the turbine pressure ratio. Thus, instead of equation (9.5) the work requirement is expressed by

$$\eta_m c_{pg} \Delta T_{034} = c_{pa} \Delta T_{012}$$

Rewriting in terms of non-dimensional groups,

$$\frac{\Delta T_{034}}{T_{03}} = \frac{\Delta T_{012}}{T_{01}} \times \frac{T_{01}}{T_{03}} \times \frac{c_{pa}}{c_{pg} \eta_m} \quad (9.6)$$

Equations (9.1), (9.2) and (9.6) are all linked by the temperature ratio T_{03}/T_{01} and it is necessary to determine, by trial and error, the turbine inlet temperature required for operation at any arbitrary point on the compressor characteristic. The procedure is as follows:

- (i) Having selected a point on the compressor characteristic, the values of $N/\sqrt{T_{01}}$, p_{02}/p_{01} , $m\sqrt{T_{01}/p_{01}}$ and η_c are determined, and $\Delta T_{012}/T_{01}$ can be calculated from equation (9.4).
- (ii) If a value of p_{03}/p_{04} is guessed, the value of $m\sqrt{T_{03}/p_{03}}$ can be obtained from the turbine characteristic, enabling the temperature ratio T_{03}/T_{01} to be obtained from the flow compatibility equation (9.2).

- (iii) This value of T_{03}/T_{01} can now be used to calculate $N/\sqrt{T_{03}}$ from equation (9.1).
- (iv) With $N/\sqrt{T_{03}}$ and p_{03}/p_{04} known, the turbine efficiency can be obtained from the turbine characteristic.
- (v) The non-dimensional temperature drop $\Delta T_{034}/T_{03}$ can be calculated from equation (9.3) and used in conjunction with equation (9.6) to calculate another value of T_{03}/T_{01} .
- (vi) This second value of T_{03}/T_{01} will not, in general, agree with the first value obtained from equation (9.2), indicating that the guessed value of the pressure ratio p_{03}/p_{04} is not valid for an equilibrium running point.
- (vii) A new value of p_{03}/p_{04} must now be assumed and the above calculations repeated until the same value of T_{03}/T_{01} is obtained from both equations (9.2) and (9.6).
- (viii) Agreement signifies that the turbine operating point is compatible with the originally chosen compressor operating point when the fuel flow is such as to produce the iterated value of T_{03}/T_{01} .

The procedure is summarized in the information flow chart of Fig. 9.6(a).

It would be possible to carry out this calculation for a large number of points on the compressor characteristic, and express the results by joining up points of constant T_{03}/T_{01} on the compressor characteristic as indicated by the dashed lines in Fig. 9.7. In practice, this is unnecessary, because the further requirement for flow compatibility with the component downstream of the gas generator, whether it be a power turbine or propelling nozzle, seriously restricts the operating zone on the compressor characteristic. The foregoing procedure thus comprises only part of the calculation for the whole unit, and a further iteration to satisfy flow compatibility with the downstream component is required, as shown in Fig. 9.6(b); this will be discussed in the next section. The free turbine engine will be considered first.

Before doing so, it is necessary to emphasize that the matching procedure just outlined has been developed on the assumption that the turbine non-dimensional flow is independent of the non-dimensional speed and is a function only of pressure ratio. If the turbine characteristic does exhibit a variation of $m\sqrt{T_{03}}/p_{03}$ with $N/\sqrt{T_{03}}$, as in Fig. 7.24, the procedure must be modified. There is no change for any turbine operating point falling on the choking point of the mass flow curve, but for others the process would be as follows. Immediately after Step (i) it is necessary to guess a value of T_{03}/T_{01} , which permits calculation of $N/\sqrt{T_{03}}$ from equation (9.1) and $m\sqrt{T_{03}}/p_{03}$ from equation (9.2). p_{03}/p_{04} and η_t can then be obtained from the turbine characteristic, enabling $\Delta T_{034}/T_{03}$ to be calculated from equation (9.3). The work compatibility equation (9.6) is then used to provide a value of T_{03}/T_{01} for comparison with the original guess. To avoid obscuring the main principles involved, the use of multi-line turbine flow characteristics will not be discussed further.[†] Enough has been said to show that they can be accommodated without undue difficulty.

[†] The characteristic shown in Fig. 7.24 is typical of that for a turbine in which choking occurs in the stator passages. In some designs choking may occur initially at exit from the rotor passages; this will result in a small variation of the choking value of $m\sqrt{T_{03}}/p_{03}$ with $N/\sqrt{T_{03}}$. The modification to the calculation would then apply also to the choking region.

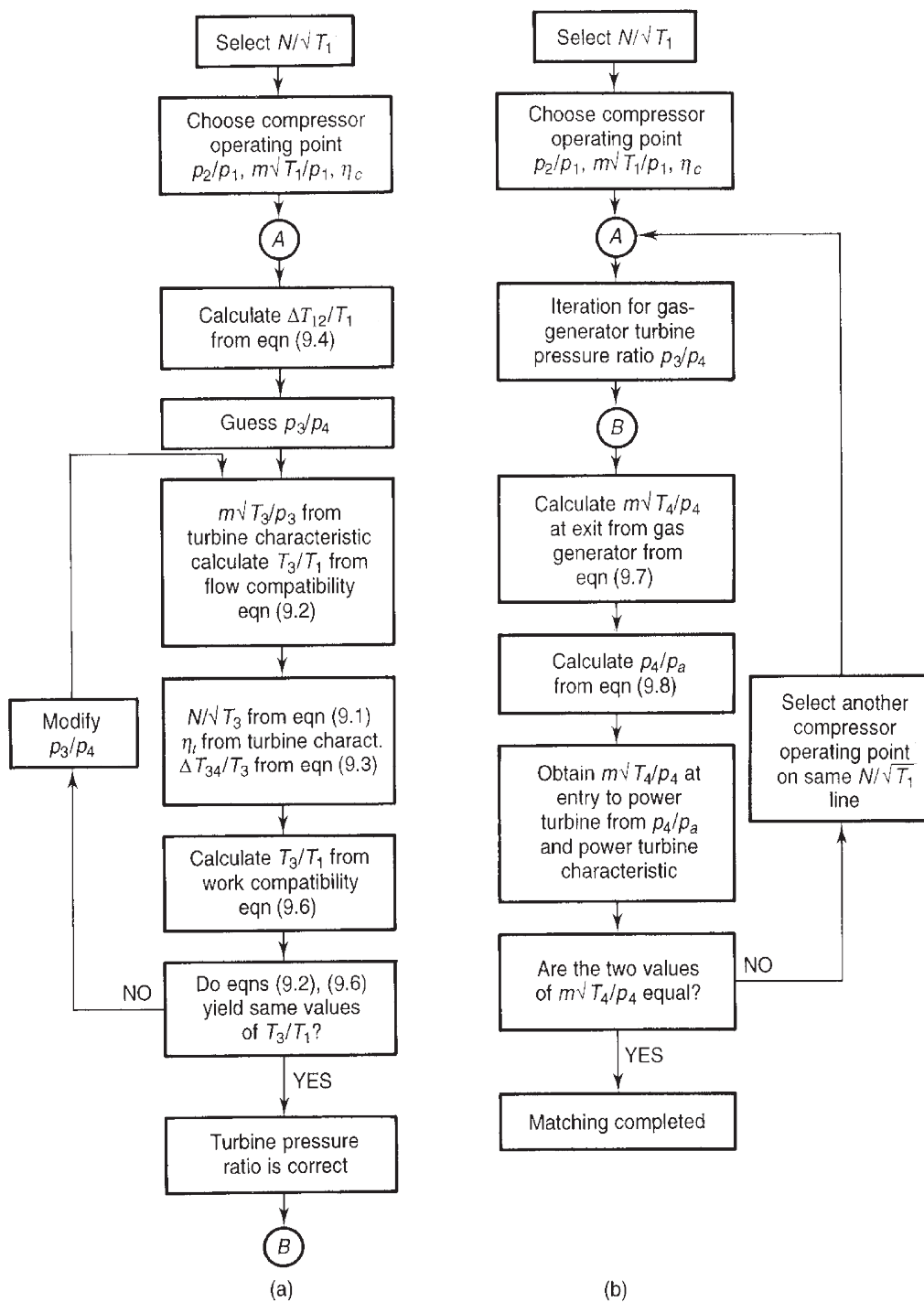


FIG. 9.6 (a) Iteration for gas generator. (b) Overall iteration procedure for free power turbine unit

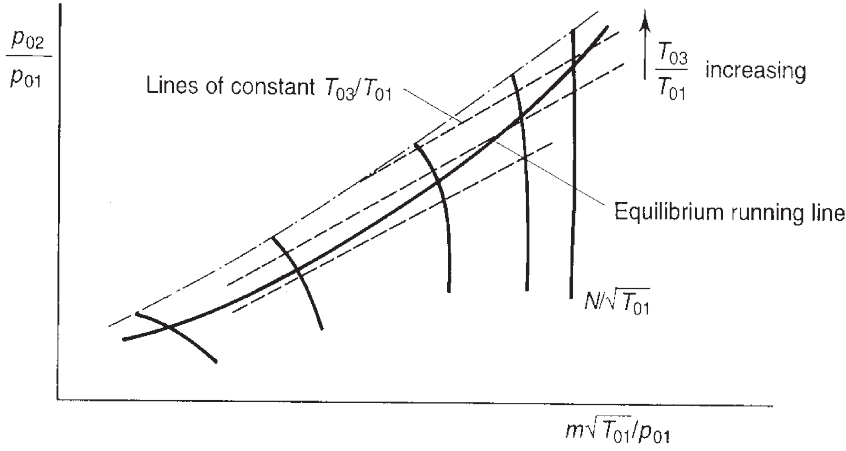


FIG. 9.7 Equilibrium running line for free turbine

9.4 Off-design operation of free turbine engine

Matching of gas generator with free turbine

The gas generator is matched to the power turbine by the fact that the mass flow leaving the gas generator must equal that at entry to the power turbine, coupled with the fact that the pressure ratio available to the power turbine is fixed by the compressor and gas-generator turbine pressure ratios. The power turbine characteristic will have the same form as Fig. 9.3, but the parameters will be $m\sqrt{T_{04}}/p_{04}$, p_{04}/p_a , N_p/T_{04} and η_{tp} .

The preceding section described how the gas-generator operating conditions can be determined for any point on the compressor characteristic. The value of $m\sqrt{T_{04}}/p_{04}$ at exit from the gas generator can then be calculated from

$$\frac{m\sqrt{T_{04}}}{p_{04}} = \frac{m\sqrt{T_{03}}}{p_{03}} \times \frac{p_{03}}{p_{04}} \times \sqrt{\frac{T_{04}}{T_{03}}} \quad (9.7)$$

where

$$\sqrt{\frac{T_{04}}{T_{03}}} = \sqrt{\left(1 - \frac{\Delta T_{034}}{T_{03}}\right)} \text{ and } \frac{\Delta T_{034}}{T_{03}} = \eta_t \left[1 - \left(\frac{1}{p_{03}/p_{04}}\right)^{(\gamma-1)/\gamma}\right]$$

The corresponding pressure ratio across the power turbine can also be established from

$$\frac{p_{04}}{p_a} = \frac{p_{02}}{p_{01}} \times \frac{p_{03}}{p_{02}} \times \frac{p_{04}}{p_{03}} \quad (9.8)$$

(It must be remembered that in the case of a stationary gas turbine with inlet and exit duct losses ignored, $p_{01} = p_a$, and the power turbine outlet pressure is also equal to p_a .)

Having found the pressure ratio across the power turbine, the value of $m\sqrt{T_{04}}/p_{04}$ can be found from the power turbine characteristic for comparison with the value obtained from equation (9.7). If agreement is not reached it is necessary to choose another point on the same constant speed line of the compressor characteristic and repeat the procedure until the requirement of flow compatibility between the two turbines is satisfied. The overall procedure for the free turbine engine, including the iteration for the gas generator, is summarized in the information flow chart of Fig. 9.6(b).

For each constant $N/\sqrt{T_{01}}$ line on the compressor characteristic there will be only one point which will satisfy both the work requirement of the gas generator and flow compatibility with the power turbine. If the foregoing calculations are carried out for each constant speed line, the points obtained can be joined up to form the equilibrium running line as shown in Fig. 9.7. *The running line for the free turbine engine is independent of the load and is determined by the swallowing capacity of the power turbine.* This is in contrast to the behaviour of the single-shaft unit, where the running line depends on the characteristic of the load as indicated in Fig. 9.5.

The interactive procedure shown in Fig. 9.6 is referred to as *nested loops* in computer terminology. It will be appreciated that as engine complicity increases, e.g. a twin-spool gas generator with a free turbine or a triple-spool turbofan, the number of nested loops increases and the method described becomes less viable. Computer-based solutions, while based on the same thermodynamic principles, take a different approach. Preliminary estimates of an operating point would be made, and these would result in errors in the compatibility requirements. For example, if the operating point on a specified speed line were to be found for a free turbine engine, the procedure could start with trial values of compressor pressure ratio, turbine inlet temperature and gas-generator pressure ratio; this would result in errors in work compatibility, gas-generator non-dimensional flow and power turbine non-dimensional flow. A Newton–Raphson iteration procedure would then be used to minimize the errors, by adjusting the estimated parameters; this would normally be done using a matrix solution for more complex engines. This method is used by most commercially available software, but does not give a ready insight into the overall matching procedure.

Having established a series of equilibrium running points, the calculation can then proceed to determine the power output and specific fuel consumption. Before discussing this, however, a useful relationship which simplifies the foregoing procedure will be introduced. It arises from the behaviour of two turbines in series, and is introduced here because it facilitates a better physical understanding of some of the phenomena discussed in subsequent sections.

Matching of two turbines in series

The iterative procedure required for the matching of a gas generator and a free turbine can be considerably simplified if the behaviour of two turbines in series is considered; this approach is also valuable for the analysis of more complex gas turbines considered in Chapter 10. It was shown in the previous sub-section that

by using equation (9.7) the value of $m\sqrt{T_{04}}/p_{04}$ at exit from the gas-generator turbine can be obtained for any gas-generator operating point, and in particular that it is a function of $m\sqrt{T_{03}}/p_{03}$, p_{03}/p_{04} and η_t . The value of η_t could be read from the turbine characteristic because $N/\sqrt{T_{03}}$ had been determined for the operating point in the course of the gas-generator calculation. Now in practice the variation of η_t at any given pressure ratio is not large (see Fig. 9.3), particularly over the restricted range of operation of the gas-generator turbine. Furthermore, such a variation has little effect on $m\sqrt{T_{04}}/p_{04}$ because the resulting change in $\sqrt{(T_{03}/T_{04})}$ is very small. It is often sufficiently accurate to take a mean value of η_t at any given pressure ratio, so that $m\sqrt{T_{04}}/p_{04}$ becomes a function only of $m\sqrt{T_{03}}/p_{03}$ and p_{03}/p_{04} . If this is done, a single curve representing the turbine outlet flow characteristic can readily be obtained by applying equation (9.7) to points on the single curve of the inlet flow characteristic. The result is shown by the dashed curve in Fig. 9.8.

The effect of operating two turbines in series is also shown in Fig. 9.8 where it can be seen that the requirement for flow compatibility between the two turbines places a major restriction on the operation of the gas-generator turbine. In particular, as long as the *power turbine is choked* the gas-generator turbine will operate at a *fixed non-dimensional point*, i.e. at the pressure ratio marked (a). With the power turbine unchoked, the gas generator will be restrained to operate at a fixed pressure ratio for each power turbine pressure ratio, e.g. (b) and (c). Thus the maximum pressure ratio across the gas-generator turbine is controlled by choking of the power turbine, and at all times the pressure ratio is controlled by the swallowing capacity of the power turbine.

A further consequence of the fixed relation between the turbine pressure ratios is that it is possible to plot the gas-generator pressure ratio p_{03}/p_{04} against compressor pressure ratio p_{02}/p_{01} by using the identity

$$\frac{p_{03}}{p_{04}} = \frac{p_{03}}{p_{02}} \times \frac{p_{02}}{p_{01}} \times \frac{p_a}{p_{04}} \quad (9.9)$$

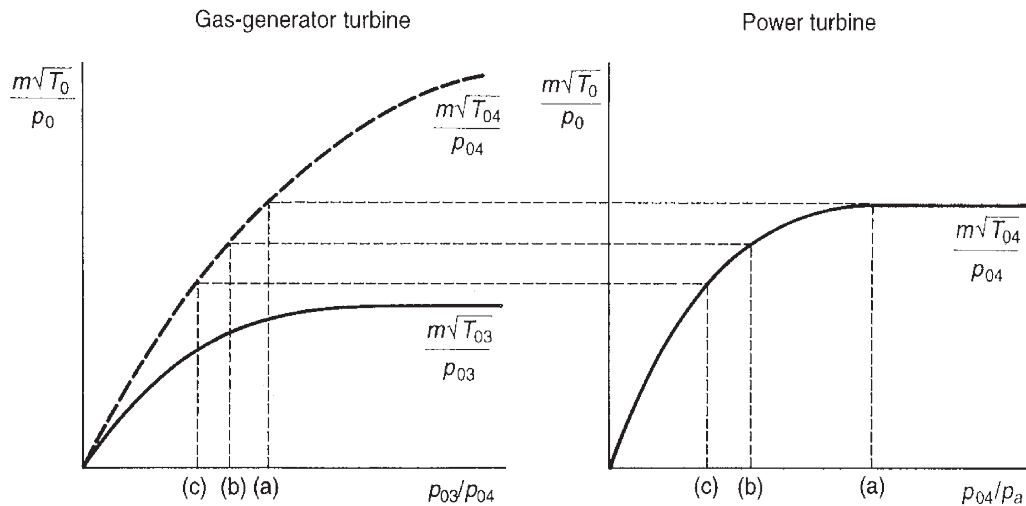


FIG. 9.8 Operation of turbines in series

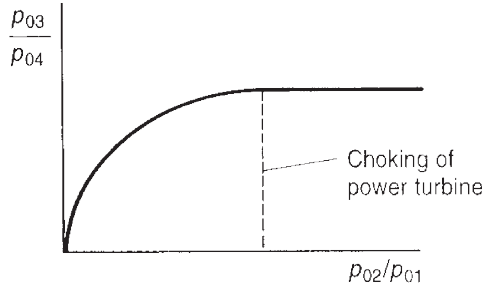


FIG. 9.9

p_{03}/p_{02} is determined by the assumed combustion pressure loss, and p_{04}/p_a is obtained from Fig. 9.8 for any value of p_{03}/p_{04} . Such a curve is shown in Fig. 9.9. From it, for any value of compressor pressure ratio, the pressure ratio of the gas-generator turbine can be determined; this in turn fixes the values of $m\sqrt{T_{03}}/p_{03}$ and $\Delta T_{034}/T_{03}$ which are required for use with equations (9.2) and (9.6). Thus it is no longer necessary to carry out the iteration for the pressure ratio of the gas-generator turbine and, for each constant speed line considered, only a *single* iteration is required to find the correct equilibrium running point.

Variation of power output and SFC with output speed

The net power output of the free turbine engine is simply the output from the power turbine, namely

$$\text{power output} = mc_p \Delta T_{045} \quad (9.10)$$

where

$$\Delta T_{045} = \eta_{tp} T_{04} \left[1 - \left(\frac{1}{p_{04}/p_a} \right)^{(\gamma-1)/\gamma} \right]$$

For each equilibrium running point established (one for each compressor speed line) p_{04}/p_a will be known, and T_{04} can be readily calculated from

$$T_{04} = T_{03} - \Delta T_{034} \quad (9.11)$$

The mass flow is obtained from $m\sqrt{T_{01}}/p_{01}$ for assumed values of p_a and T_a . The outstanding unknown is the power turbine efficiency η_{tp} . It can be obtained from the power turbine characteristic, but it depends not only on the pressure ratio p_{04}/p_a but also on $N_p/\sqrt{T_{04}}$, i.e. on the power turbine speed N_p . Free turbine engines are used to drive a variety of loads such as pumps, propellers and electric generators, each with a different power versus speed characteristic. For this reason it is usual to calculate the power output over a range of power turbine speed for each equilibrium running point (i.e. for each compressor speed). The results could be plotted as in Fig. 9.10. Any curve corresponding to a given compressor speed will

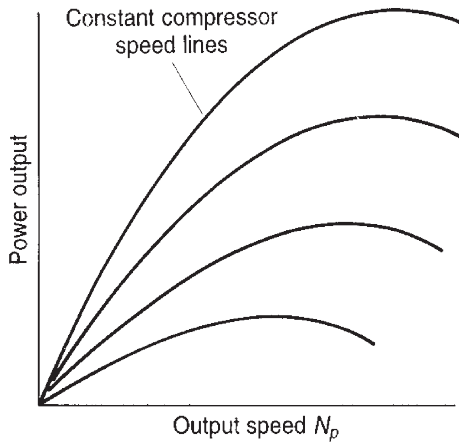


FIG. 9.10

be fairly flat in the useful upper half of the output speed range where η_{tp} does not vary much with $N_p/\sqrt{T_{04}}$ (see Fig. 9.3).

The fuel consumption can also be calculated for each equilibrium running point, in the same manner as described in section 9.2 for the single-shaft unit. Since it depends only on the gas-generator parameters, there will be only one value for each compressor speed. When combined with the power output data to give the *SFC*, however, it is clear that the *SFC*, like the power output, will be a function of both compressor speed and power turbine speed. It is convenient to express the off-design performance by plotting *SFC* against power output for several power turbine speeds as shown in Fig. 9.11. This type of presentation permits the customer to evaluate the performance of the unit when coupled to the specified load by superimposing the load characteristic upon it. The dashed curve in Fig. 9.11 indicates a particular variation of power and speed imposed by a load, and the points of intersection with the N_p curves give the *SFC* versus power output curve for the free turbine engine driving that particular load. Figure 9.11 relates to operation at one ambient condition, and normally the performance calculations would be repeated for extreme values of p_a and T_a .

It should be appreciated that although for convenience of calculation the compressor speed has been chosen as the independent variable, in practice the *fuel flow* is the independent variable. A chosen value of fuel flow (and hence T_{03}) determines the compressor speed and ultimately the power output. The *SFC* curves of Fig. 9.11 exhibit an increase in *SFC* as the power is reduced because the reduction in fuel flow leads to a reduction in compressor speed and gas-generator turbine inlet temperature. It will be remembered from Chapter 2 that the efficiency of a real cycle falls as the turbine inlet temperature is reduced. This poor part-load economy is a major disadvantage of the simple gas turbine. Consideration of the substantial improvement arising from the use of more complex cycles will be deferred until Chapter 9. Referring again to Fig. 9.11, the change in *SFC* with N_p at any given power output is not very marked, because the gas-generator parameters change only slightly under this operating condition.

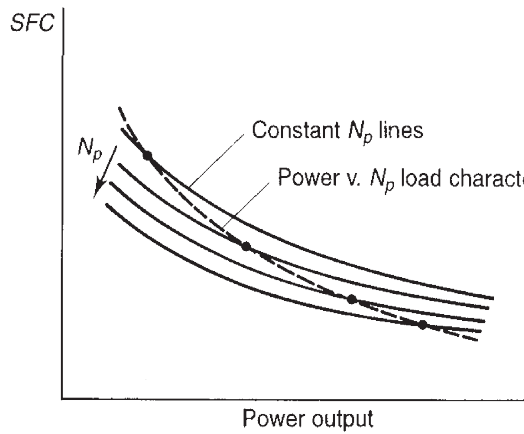


FIG. 9.11

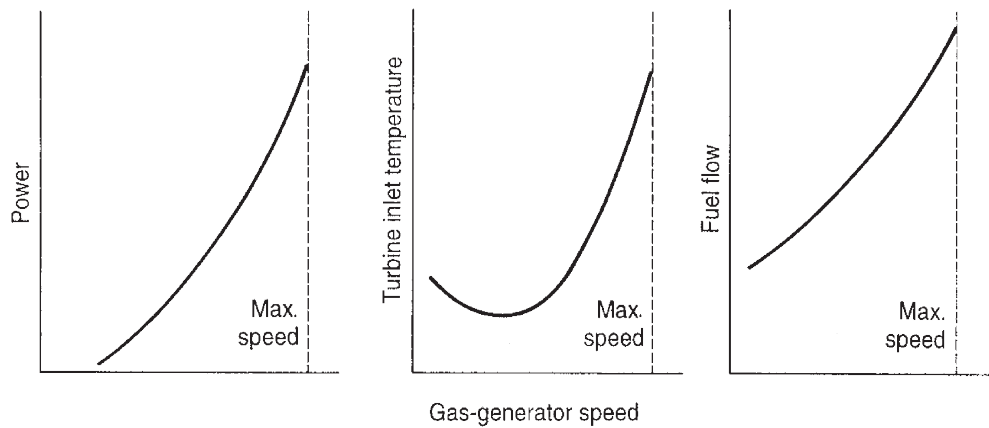


FIG. 9.12 Variation of key parameters with gas-generator speed

It is useful to consider the variation of the key variables (power, turbine inlet temperature and fuel flow) with gas-generator speed as shown in Fig. 9.12. It can be seen that all increase rapidly as the gas generator approaches its maximum permissible speed. The change in turbine inlet temperature is especially critical because of the effect on creep life of the first-stage turbine blades and, in general, operation at the maximum speed would only be for limited periods. In electricity generation, for example, this maximum speed would be used for emergency peak duty, and base-load operation would utilize a reduced gas-generator speed with correspondingly increased blade life.

Single-shaft versus twin-shaft engines

The choice of whether to use a single-shaft or twin-shaft (free turbine) power plant is largely determined by the characteristics of the driven load. An electric generator requires a constant rotational speed and an engine designed specifically for this

application would make use of a single-shaft configuration (see Fig. 1.24). An alternative, however, is the use of an aircraft derivative with a free power turbine in the place of the propelling nozzle. With this arrangement it is possible to design a power turbine of substantially larger diameter than the gas generator, using an elongated duct between the gas generator and the power turbine; this then permits the power turbine to operate at the required electric generator speed without the need for a reduction gearbox. Turboprops may use either configuration, as shown in Figs 1.10 and 1.11.

The running lines for single-shaft and twin-shaft units were shown in Figs 9.5 and 9.7. It should be noted that in the case of the single-shaft engine driving a generator, reduction in output power results in a slight increase in compressor mass flow; although there is some reduction in compressor pressure ratio, there is little change in compressor temperature rise because the efficiency is also reduced. This means that the compressor power remains essentially fixed. With a twin-shaft engine, however, as Fig. 9.7 shows, reducing net power output involves a reduction in compressor speed and hence in airflow, pressure ratio and temperature rise. The compressor power needed is therefore appreciably lower than for the single-shaft engine. It should also be evident from a comparison of Figs 9.5 and 9.7 that the compressor operates over a smaller range of efficiency in a twin-shaft engine. For these reasons the part-load fuel consumption of a twin-shaft engine is superior when driving a constant speed load. Performance calculations for a 1960s vintage engine available in both single-shaft and two-shaft versions showed this to be true. Similar calculations for a 1990s vintage engine by the same manufacturer, however, showed virtually identical *SFC* across the running range. The reason for the difference between old and current engines is that the 1960s engine had no variable geometry, while the 1990s engine used variable geometry in both configurations. The single-shaft version used variable IGVs to maintain high exhaust temperature as load was reduced, with a consequent reduction in compressor mass flow and power, thus improving *SFC* at lower power settings. The two-shaft version, on the other hand, used compressor blow-off at reduced power as part of the low-emissions combustion strategy, resulting in a penalty in low-load *SFC*.

The two types also have different characteristics regarding the supply of waste heat to a cogeneration plant, primarily due to the differences in exhaust flow as load is reduced; the essentially constant airflow and compressor power in a single-shaft unit results in a larger decrease of exhaust temperature for a given reduction in power, which might necessitate the burning of supplementary fuel in the waste heat boiler under operating conditions where it would be unnecessary with a twin-shaft unit. In both cases, the exhaust temperature may be increased by the use of variable IGVs. Cogeneration systems have been successfully built using both single-shaft and twin-shaft units, the latter often being aircraft derivative engines.

Most gas turbines do not spend long periods at low power and the *SFC* differences may not be critical in practice. The *torque characteristics*, however, are very different and the variation of torque with output speed at a given power may well determine the engine's suitability for certain applications, e.g. a high starting torque is particularly important for traction purposes.

The compressor of a *single-shaft engine* is constrained to turn at some multiple of the load speed, fixed by the transmission gear ratio, so that a wide range of load speed implies a reduction in compressor speed. This results in a reduction in mass flow and hence of output torque as shown by curve (a) of Fig. 9.13. This type of turbine is clearly unsuitable for traction purposes. The normal flat torque curve of an internal combustion engine is shown dashed for comparison.

The *free power turbine unit*, however, has a torque characteristic even more favourable than the internal combustion engine. The variation of power output with load speed, at any given compressor speed determined by the fuel flow, is shown in Fig. 9.10. It can be seen that the output power remains relatively constant over a wide load speed range for a fixed compressor speed. This is due to the fact that the compressor can supply an essentially constant flow at a given compressor speed, irrespective of the free turbine speed. Thus at fixed gas-generator operating conditions, reduction in output speed results in an increase in torque as shown by curve (b) in Fig. 9.13. It is quite possible to obtain a stall torque of two to three times the torque delivered at full speed.

The actual range of speed over which the torque conversion is efficient depends on the efficiency characteristic of the power turbine. The typical turbine efficiency characteristic shown in Fig. 9.3 suggests that the fall in efficiency will not be greater than about 5 or 6 per cent over a speed range from half to full speed. Thus quite a large increase in torque can be obtained efficiently when the output speed is reduced to 50 per cent of its maximum value. The efficiency of the torque conversion at low speeds, e.g. when accelerating a vehicle from rest, will be very low. In some applications a simple two-speed gearbox might be sufficient to overcome this defect, but it is probable that gas turbines for heavy road vehicles will incorporate some form of automatic transmission with five or six speeds.

Further differences between single- and twin-shaft units are brought out in section 10.5 where the transient behaviour of gas turbines is discussed.

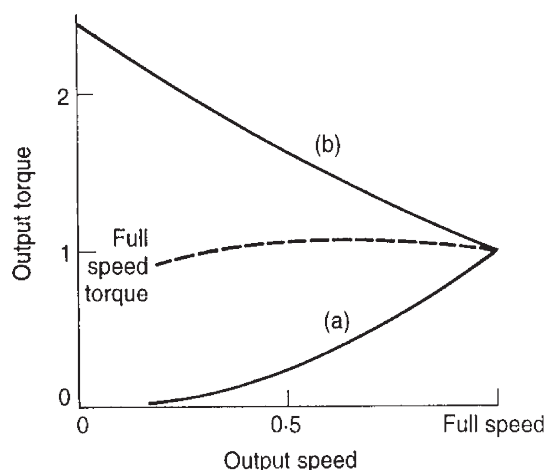


FIG. 9.13 Torque characteristics

EXAMPLE 9.2

A gas turbine with a free power turbine is designed for the following conditions:

Air mass flow	30 kg/s
Compressor pressure ratio	6.0
Compressor isentropic efficiency	0.84
Turbine inlet temperature	1200 K
Turbine isentropic efficiency (both turbines)	0.87
Combustion pressure loss	0.20 bar
Mechanical efficiency (applies to gas generator and load)	0.99
Ambient conditions	1.01 bar, 288 K

Calculate the power developed and the turbine non-dimensional flows $m\sqrt{T_{03}/p_{03}}$ and $m\sqrt{T_{04}/p_{04}}$ at the design point.

If the engine is run at the same mechanical speed at an ambient temperature of 268 K calculate the values of turbine inlet temperature, pressure ratio and power output, assuming the following:

- combustion pressure loss remains constant at 0.20 bar;
- both turbines choking, with values of $m\sqrt{T_{03}/p_{03}}$ and $m\sqrt{T_{04}/p_{04}}$ as calculated above and no change in turbine efficiency;
- at 268 K and the same N , the $N/\sqrt{T_{01}}$ line on the compressor characteristic is a vertical line with a non-dimensional flow 5 per cent greater than the design value;
- variation of compressor efficiency with pressure ratio at the relevant value of $N/\sqrt{T_{01}}$ is

p_{02}/p_{01}	6.0	6.2	6.4	6.6
η_c	0.837	0.843	0.845	0.840

The design point calculation is straightforward and only the salient results are presented:

Gas-generator turbine	
pressure ratio	2.373
inlet pressure	5.86 bar
temperature drop	203 K
Power turbine	
pressure ratio	2.442
inlet pressure	2.47 bar
temperature drop	173.5 K
inlet temperature	997 K

The power output is then $(30 \times 1.148 \times 173.5 \times 0.99)$ kW or 5910 kW.

The design point values of $m\sqrt{T_{03}}/p_{03}$ and $m\sqrt{T_{04}}/p_{04}$ are

$$\frac{30 \times \sqrt{1200}}{5.86} 177.4 \text{ and } \frac{30 \times \sqrt{997}}{2.47} = 383.5 \text{ respectively}$$

At 268 K the value of $m\sqrt{T_{01}}/p_{01}$ is

$$1.05 \times \left[\frac{30 \times \sqrt{288}}{1.01} \right] 529.5$$

If the power turbine remains choked, the gas-generator turbine will be constrained to operate at a fixed non-dimensional point, and thus the value of $\Delta T_{034}/T_{03} = 203/1200 = 0.169$, as for the design condition.

For work compatibility

$$\frac{\Delta T_{012}}{T_{01}} = \frac{\Delta T_{034}}{T_{03}} \times \frac{T_{03}}{T_{01}} \left[\frac{c_{pg}\eta_m}{c_{pa}} \right] = \frac{0.169 \times 1.148 \times 0.99}{1.005} \times \frac{T_{03}}{T_{01}}$$

and hence

$$\frac{T_{03}}{T_{01}} = 5.23 \frac{\Delta T_{012}}{T_{01}} \quad (\text{A})$$

For flow compatibility

$$\begin{aligned} \frac{m\sqrt{T_{03}}}{p_{03}} &= \frac{m\sqrt{T_{01}}}{p_{01}} \times \frac{p_{01}}{p_{03}} \times \sqrt{\frac{T_{03}}{T_{01}}} \\ 177.4 &= 529.5 \frac{p_{01}}{p_{03}} \times \sqrt{\frac{T_{03}}{T_{01}}} \\ \sqrt{\frac{T_{03}}{T_{01}}} &= 0.335 \frac{p_{03}}{p_{01}} \end{aligned} \quad (\text{B})$$

The problem is thus to find the compressor operating point that will give the same value of T_{03}/T_{01} for both equations (A) and (B). With the variation in efficiency prescribed, the value of $\Delta T_{012}/T_{01}$ can readily be calculated from equation (9.4), and with the constant value of combustion pressure loss the value of p_{03} can also be calculated. The solution is best carried out in tabular form as shown:

$\frac{p_{02}}{p_{01}}$	$\left[\left(\frac{p_{02}}{p_{01}} \right)^{(\gamma-1)/\gamma} - 1 \right]$	η_c	$\frac{\Delta T_{012}}{T_{01}}$	$\left(\frac{T_{03}}{T_{01}} \right)_A$	p_{02}	p_{03}	$\frac{p_{03}}{p_{01}}$	$\sqrt{\frac{T_{03}}{T_{01}}}$	$\left(\frac{T_{03}}{T_{01}} \right)_B$
6.0	0.669	0.837	0.799	4.18	6.06	5.86	5.80	1.943	3.78
6.2	0.684	0.843	0.812	4.25	6.26	6.06	6.00	2.010	4.05
6.4	0.700	0.845	0.828	4.33	6.46	6.26	6.20	2.078	4.32
6.6	0.715	0.840	0.851	4.45	6.66	6.46	6.40	2.144	4.60

Solving graphically, the required pressure ratio is found to be 6.41, with the value of $T_{03}/T_{01} = 4.34$; the corresponding turbine inlet temperature is $4.34 \times 268 = 1163$ K.

Having established the compressor pressure ratio and turbine inlet temperature, it is a straightforward matter to calculate the power developed. Remembering that the gas-generator turbine will still operate at the same non-dimensional point ($\Delta T_{034}/T_{03} = 0.169$, $p_{03}/p_{04} = 2.373$) the power turbine entry conditions and temperature drop can readily be calculated; the resulting temperature drop is 179.6 K. The mass flow is obtained from the non-dimensional flow and the ambient conditions:

$$\frac{m\sqrt{T_{01}}}{p_{01}} = 529.5, \text{ hence } m = \frac{529.5 \times 1.01}{\sqrt{268}} = 32.7 \text{ kg/s}$$

and power output = $32.7 \times 1.148 \times 179.6 \times 0.99 = 6680 \text{ kW}$.

From this example, it can be seen that operation at the design mechanical speed on a cold day results in a decrease of maximum cycle temperature from 1200 to 1163 K, even though the value of T_{03}/T_{01} has increased from 4.17 to 4.34 due to the increase in $N/\sqrt{T_{01}}$. The power has increased from 5910 to 6680 kW and this can be seen to be due to the simultaneous increase in air mass flow and overall pressure ratio. The beneficial effect of low ambient temperature on gas turbine operation is evident; conversely, high ambient temperatures result in significant penalties. The effect of increased ambient temperature on turbojet operation is discussed in the next section under the heading 'Variation of thrust with rotational speed, forward speed and altitude'.

Industrial gas turbines used to generate electricity in regions with high summer temperatures may have to meet very high peak demands due to large air-conditioning loads for about 25 per cent of the year. It is possible to increase power by cooling the air entering the compressor, and electricity utilities have found this to be an economical solution compared with the alternative of buying a more powerful gas turbine which would need to run at reduced power and efficiency for 75 per cent of the year. Several different methods have been used. In regions of low relative humidity, cooling can be achieved using an *evaporative cooler*; the incoming air is passed through a wetted filter, and the heat required to evaporate the water leads to a reduction in temperature. This approach, however, is not effective in areas of high relative humidity. An alternative approach, successfully introduced in the mid 1990s, is *ice harvesting*. At off-peak times, primarily overnight, electricity is used to drive refrigeration chillers which produce and store large quantities of ice. When maximum power is needed during the heat of the day, ice is melted and the chilled water is used in heat-exchangers to lower the inlet temperature. This method has proved to be economically attractive, producing substantial increases in power at a low cost per kilowatt [Ref. (1)]. Ice harvesting, however, requires construction of large ice storage containers and a new approach, referred to as *fog cooling*, was introduced in the late 1990s. The 'fog' is produced by injecting finely atomized water droplets, forced through tiny nozzles at pressures up to 200 bar, into the inlet air stream. The use of very small droplets (5–10 microns), in very large quantities, provides a large surface area for evaporation

and cooling of the intake air; this approach is clearly superior to evaporative cooling in regions of high relative humidity and any excess water not evaporated will give an increase in mass flow, resulting in more power. It is important that water brought into the airflow is demineralized, to prevent the build-up of deposits on the blading causing performance deterioration due to loss of compressor efficiency. Reference (2) gives a good review of the advantages of inlet cooling.

9.5 Off-design operation of the jet engine

Propelling nozzle characteristics

The propelling nozzle area for a jet engine is determined from the design point calculations as described in Chapter 3, and once the nozzle size has been fixed it has a major influence on the off-design operation. The characteristic for a nozzle, in terms of 'non-dimensional' flow $m\sqrt{T_{04}}/p_{04}$ and pressure ratio p_{04}/p_{05} , can readily be calculated as follows.

The mass flow parameter is given by

$$\begin{aligned}\frac{m\sqrt{T_{04}}}{p_{04}} &= C_5 A_5 \rho_5 \frac{\sqrt{T_{04}}}{p_{04}} \\ &= \frac{C_5}{\sqrt{T_{04}}} \times \frac{A_5}{R} \times \frac{p_5}{p_{04}} \times \frac{T_{04}}{T_5}\end{aligned}\quad (9.12)$$

where A_5 is the effective nozzle area. Making use of equation (3.12), $C_5/\sqrt{T_{04}}$ can be found from

$$\frac{C_5^2}{T_{04}} = 2c_p \eta_j \left[1 - \left(\frac{1}{p_{04}/p_5} \right)^{(\gamma-1)/\gamma} \right] \quad (9.13)$$

and T_5/T_{04} from

$$\frac{T_5}{T_{04}} = 1 - \frac{T_{04} - T_5}{T_{04}} = 1 - \eta_j \left[1 - \left(\frac{1}{p_{04}/p_5} \right)^{(\gamma-1)/\gamma} \right] \quad (9.14)$$

It follows that for a nozzle of given area and efficiency, $m\sqrt{T_{04}}/p_{04}$ can be calculated as a function of the pressure ratio p_{04}/p_5 . But equations (9.13) and (9.14) are valid for pressure ratios only up to the critical value, given by equation (3.14), namely

$$\frac{p_{04}}{p_c} = 1 / \left[1 - \frac{1}{\eta_j} \left(\frac{\gamma-1}{\gamma+1} \right) \right]^{\gamma/(\gamma-1)} \quad (9.15)$$

and up to this point $p_5 = p_a$. For pressure ratios p_{04}/p_a greater than the critical, $m\sqrt{T_{04}}/p_{04}$ remains constant at the maximum (choking) value, i.e. it is independent of p_{04}/p_a (and incidentally $p_5 = p_c > p_a$). Thus $m\sqrt{T_{04}}/p_{04}$ can be plotted versus